Math 526, F06 **Name:**
Quiz 4 **SSN: xxx-xx-**

<u>Instructions:</u> **There are totally 1 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

Problem 1. (20 points) Let $\mathbf{A} = \begin{bmatrix} 0 & 2 & -2 \\ 1 & -1 & 1 \\ -1 & 3 & -1 \end{bmatrix}$.

a) Prove that $\mathbf{A}$ is invertible.

Proof: We first do the $\mathbf{L}-\mathbf{U}-\mathbf{P}$ factorization of $\mathbf{A}$.

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \quad \begin{bmatrix} 0 & 2 & -2 \\ 1 & -1 & 1 \\ -1 & 3 & -1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \quad \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -2 \\ -1 & 3 & -1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} & & \\ 0 & & \\ -1 & & \end{bmatrix} \quad \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -2 \\ 0 & 2 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} & & \\ 0 & & \\ -1 & 1 & \end{bmatrix} \quad \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 2 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So we have

$$\mathbf{L} = \begin{bmatrix} 1 & & \\ 0 & 1 & \\ -1 & 1 & 1 \end{bmatrix}, \qquad \mathbf{U} = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 2 \end{bmatrix}, \qquad \mathbf{P} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Since all diagonal components of $\mathbf{U}$ are non-zero, we conclude that $\mathbf{U}$ is invertible and consequently $\mathbf{A}$ is also invertible.

b) Find $\mathbf{A}^{-1}$.

Solution: The first column of A^{-1} is computed by solving

$$\mathbf{A}\mathbf{x} = \mathbf{e}_1$$

i.e.

$$\begin{bmatrix} 0 & 2 & -2 \\ 1 & -1 & 1 \\ -1 & 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Using the $\mathbf{L}-\mathbf{U}-\mathbf{P}$ factorization found in a), it is easy to get

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 0 \\ -1/2 \end{bmatrix} \tag{1}$$

The second column of A^{-1} is computed by solving

$$\begin{bmatrix} 0 & 2 & -2 \\ 1 & -1 & 1 \\ -1 & 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

we again get

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1/2 \\ 1/2 \end{bmatrix} \tag{2}$$

The third column of A^{-1} is computed by solving

$$\begin{bmatrix} 0 & 2 & -2 \\ 1 & -1 & 1 \\ -1 & 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

we again get

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1/2 \\ 1/2 \end{bmatrix} \tag{3}$$

Finally, we have

$$\mathbf{A}^{-1} = \begin{bmatrix} 1/2 & 1 & 0 \\ 0 & 1/2 & 1/2 \\ -1/2 & 1/2 & 1/2 \end{bmatrix}.$$